Deep Learning for Panoramic Dental Radiographs: A Comprehensive Survey on Caries Detection, Periodontal Bone Loss Assessment and Explainable AI

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***Abstract*—Dental panoramic radiographs (OPGs) are widely used for the diagnosis of dental caries and periodontal disease, but manual interpretation is still time-consuming and can vary between examiners. Although deep learning has improved significantly, current research remains fragmented, with most studies focusing on isolated tasks and lacking clinically integrated solutions.**

In this survey, deep learning methods for panoramic dental radiograph analysis are reviewed, covering tasks such as caries detection, severity staging, periodontal bone loss assessment, tooth segmentation, and explainability. We analyze 47 studies published between 2021 and 2026 using architectures including CNNs, YOLO-based detectors, U-Net variants, and transformer-based models. The work also proposes a unified multi-task dental AI framework that combines detection, localization, severity staging, quantification, and explainability within a single system.

We observe that while individual tasks achieve high performance, no existing approach provides a complete and clinically usable pipeline. Five major gaps are identified, including task fragmentation, lack of severity staging, limited explainability adoption, dataset bias, and weak clinical integration.

Overall, this survey provides a basis for developing next-generation, clinically deployable dental AI systems.

***Index Terms*—Deep Learning, Dental Caries, Periodontal Disease, Panoramic Radiographs, Tooth Segmentation and Detection, Multi-Task Learning, Explainable AI**

1. Introduction

Dental panoramic radiography, also known as Orthopantomography (OPG), is widely used in routine dental practice. A single panoramic image captures the full dentition along with the surrounding alveolar bone and anatomical landmarks. This makes it a commonly used imaging method for detecting dental caries, periodontal bone loss, impacted teeth, and other oral pathologies. Even though it is widely used, accurate interpretation of panoramic radiographs requires significant expertise and can vary between examiners, particularly among early-stage practitioners such as dental students and interns, who are more prone to diagnostic error.

Dental caries, a condition involving bacterial demineralization of tooth enamel and dentine, affects approximately 2.3 billion people globally and is the most prevalent non-communicable disease worldwide [1]. Periodontal disease, which involves irreversible alveolar bone loss around the teeth, affects over 19% of the adult global population and is the leading cause of adult tooth loss [2]. Both conditions share a key characteristic: their early radiographic signs are subtle and often missed, especially in early stages. A faint radiolucency at the proximal surface of a premolar, or a 1–2 mm reduction in alveolar bone height can be clinically important but difficult to identify, even for experienced examiners.

In recent years, deep learning has helped improve automated dental radiograph analysis. Several models are used, including convolutional neural networks (CNNs), YOLO-based object detectors, U-Net architectures, and transformer-based approaches, which have shown strong performance in tasks like caries detection, tooth segmentation, and periodontal bone loss estimation [3]–[5].

However, research in this area is still fragmented at the system level. Existing approaches predominantly focus on isolated tasks or single disease objectives, without integrating these components into a unified diagnostic workflow required in clinical practice. This shows a key limitation, where models focus on individual tasks rather than complete diagnostic workflows [6], [7].

Automated dental radiograph analysis is a complex problem involving multiple steps. A clinically useful system needs to handle several requirements at the same time. These include detecting multiple teeth, identifying low-contrast early-stage lesions, and measuring anatomical landmarks such as the cemento-enamel junction (CEJ) and alveolar bone crest, as well as the coexistence of multiple diseases within a single panoramic image. This multi-stage diagnostic process can be conceptualized as a pipeline, as illustrated in Fig. 1.

Furthermore, the lack of standardized datasets and evaluation protocols limits reproducibility and comparison across studies. Many studies rely on private, single-center datasets, leading to dataset bias and poor generalization to external clin-

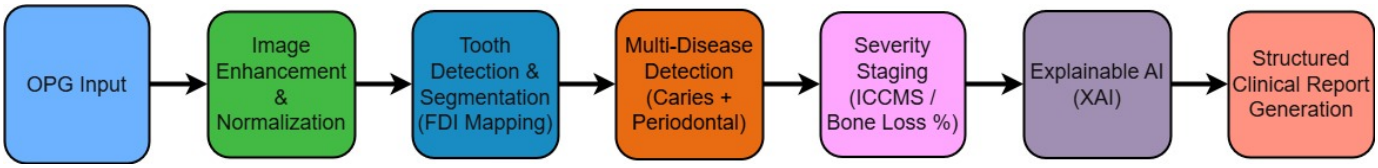


Fig. 1. System Pipeline of Proposed AI Model.

ical environments [7], [8]. Explainable artificial intelligence (XAI) is still rarely used, even though it is important for clinical trust, interpretability, and regulatory acceptance [9], [10]. The absence of explainability, combined with inconsistent validation practices, creates a major barrier to real-world deployment.

Unlike previous surveys that focus on individual tasks or models, this work takes a system-level perspective, analyzing how disparate components can be integrated into a unified diagnostic framework [11], [12].

We review 47 research papers published between 2021 and 2026 to provide a combined view of deep learning methods applied to dental radiograph analysis. The literature is organized into six categories: caries detection and severity classification, periodontal disease and bone loss analysis, multi-task end-to-end diagnosis, explainable AI, dataset benchmarking and clinical validation, and tooth segmentation and numbering. A key finding is that no existing system simultaneously addresses caries detection, severity staging, periodontal bone loss quantification, and explainability within a single unified framework for panoramic radiographs.

The main contributions are as follows:

- A task-level taxonomy of deep learning methods for panoramic dental radiograph analysis, covering detection, segmentation, classification, and multi-task architectures.
- A problem formulation for unified multi-disease dental diagnosis, highlighting the integration requirements of detection, severity staging, quantification, and explainability.
- A comparative analysis of 47 studies, identifying inconsistencies in datasets, evaluation metrics, and validation protocols.
- A quantified gap analysis highlighting key limitations in current research, including single-disease focus, lack of automated severity staging, dataset fragmentation, limited use of explainable AI.
- A research direction toward clinically deployable, explainable, multi-task dental AI systems.

Overall ,this study highlights unified, explainable, multi-task dental AI as an important next step toward clinically deployable diagnostic systems, especially for supporting junior clinicians and dental students .

2. Search Methodology

This survey used a structured keyword search which was conducted across five digital libraries: IEEE Xplore, PubMed/MEDLINE, Scopus, Google Scholar, and SpringerLink. The search was executed between January 2024 and March 2025, with publication year filters restricting results to 2020 onward.

2.1 Search Strategy

Search queries were organized into five thematic clusters:

- *Caries detection on panoramic radiographs:* “dental caries” panoramic radiograph deep learning; “caries detection” panoramic radiograph CNN; “caries segmentation” orthopantomogram neural network.
- *Periodontal disease and bone loss:* “periodontal disease” panoramic radiograph deep learning; “alveolar bone loss” orthopantomogram AI; “CEJ” “bone crest” panoramic radiograph segmentation.
- *Multi-disease and end-to-end OPG systems:* “multi-task” dental panoramic radiograph deep learning; “dental disease detection” panoramic radiograph end-to-end.
- *Explainable AI in dental imaging:* “explainable AI” dental radiograph; “Grad-CAM” dental panoramic; “interpretable deep learning” dental imaging.
- *Dataset and clinical validation:* “dental panoramic radiograph dataset” deep learning; “OPG benchmark” AI; “clinical validation” dental AI panoramic.

2.2 Inclusion and Exclusion Criteria

Papers were included if they: (i) applied deep learning or machine learning to dental radiograph analysis; (ii) used panoramic, bitewing, or periapical radiographs as primary input; (iii) targeted detection, segmentation, classification, or staging of dental caries, periodontal disease, or related pathologies; and (iv) reported quantitative performance metrics. Papers were excluded if they focused exclusively on CBCT/3D imaging, addressed only non-dental imaging, or did not report reproducible quantitative results. Clinical reviews without AI models were included only when they provided foundational clinical definitions directly relevant to the survey scope.

2.3 Final Corpus

After applying the criteria, 47 papers were selected, spanning 2021–2026, with 25 papers from 2025 (53%), 17 from 2024 (36%), 2 from 2026, 2 from 2023, and 1 from 2021 – reflecting the field’s rapid recent growth. Publication venues include BMC Oral Health (8 papers), IEEE (8), MDPI journals (9), Elsevier (4), Springer (3), and Scientific Reports/Nature Portfolio (3).

3. System-Level Taxonomy and Problem Formulation

3.1 Formal Problem Definition

Panoramic dental radiograph analysis is inherently a multi-component diagnostic problem that cannot be reduced to single-task prediction. In clinical practice, meaningful diagnosis requires coordinated execution of multiple interdependent processes: anatomical localization of teeth, identification of pathological conditions, assessment of disease severity, and quantitative measurement of clinically relevant parameters. Most existing deep learning systems in the reviewed literature address only one component in isolation [1], [6], [12], producing outputs that are diagnostically incomplete and clinically difficult to integrate.

A unified dental AI system maps an input OPG to structured diagnostic outputs through six core functional components: detection, localization, severity staging, quantification, explainability, and clinical integration. From the reviewed literature, no study in the group of 47 papers addresses all six components simultaneously on panoramic radiographs, identifying a key architectural gap that motivates this survey.

These six core components are defined below:

- **Detection:** Refers to identifying pathological conditions from radiographic features such as carious lesions, periodontal bone loss, periapical pathology, impacted teeth. YOLO-based architectures are used for this task [3], [13]–[16].
- **Localization:** refers to identifying the position of affected teeth using bounding boxes or segmentation boxes. Tooth level localization using FDI numbering is present in pipeline systems [4], [14]–[16] but absent in classification-only approaches [17]–[19].
- **Severity Staging:** Classification of disease progression along clinically meaningful scales. For caries: enamel, dentine, pulp involvement [15], [20]. For periodontal disease: Stages I–IV per 2018 AAP/EFP classification [21]. This is the most critically under addressed component, with only two C1 papers providing caries severity differentiation [15], [20].
- **Quantification:** Measurement of the CEJ–alveolar bone crest distance and percentage bone loss. Only three studies address this: Jundaeng et al. achieved mAP50 = 0.995 [4]; Wimalasiri et al. reported ICC = 0.80 [22]; and Tian et al. achieved ICC = 0.83 [23].
- **Explainability:** Generation of interpretable visual outputs for clinical verification. Only 3 of 47 papers implement explainability (6.4%): Chen et al. [24] and Bilal et al. [10] use Grad-CAM, and Chen et al. [14] show feature map attention.
- **Clinical Integration:** Alignment with diagnostic workflows, reporting standards, and regulatory requirements. Only Schwendicke et al. [25] model economic impact, and only Lee et al. [26] measure clinician acceptance through a 65-dentist usability survey.

This six-component formulation stands in direct contrast to existing approaches that address each task in isolation – a fragmentation independently identified as critical by multiple systematic reviews [2], [6], [11], [12], [27], [28].

3.2 System Taxonomy

Existing approaches are grouped into four system archetypes based on task integration and diagnostic scope. This taxonomy shows both progress toward integration and the limitations that fully prevent full clinical use. Table I summarizes the capability coverage across these categories.

1) Single-Task Approaches: Single-task systems focus on one diagnostic objective, such as, caries detection, tooth segmentation, or periodontal classification – without integration. This accounts for most reviewed studies (18 C1, 15 C2 papers). While several studies report high performance, such as CariesXrays FPCL reaching AP50 = 84.1% [14], Attention U-Net achieving Recall = 0.96 [16], U-Net variants reporting Dice = 96.77% [29] – none produce severity grading, periodontal assessment, or clinical explanations. The main limitation is not performance but scope, as these models focus on a single task without considering interactions with other diagnostic tasks. This is also supported by systematic reviews [6], [12], which highlights the lack of multi-task integration as the key limitation.

2) Multi-Task Approaches: Seven studies explore multi-task designs with varying completeness. Chen et al. [24] is one of the more complete approaches, classifying both periodontitis and caries on a per-tooth basis using EfficientNetB0 with YOLOv7 localization (ROC AUC = 98.67% for periodontitis, 98.31% for caries) with Grad-CAM explainability. However, this system operates exclusively on periapical radiographs – not panoramic OPGs – which limits its direct applicability to panoramic radiographs. Multi-task approaches shows a clear performance difference: structural tasks (such as segmentation and enumeration) achieve high accuracy, while pathological tasks (such as disease detection and severity staging) remain more challenging. Mureşanu et al. [7] reported that: YOLOv8 F1 dropped from 0.60 (internal) to 0.47 (external) – a 22% decrease due to single-center training.

3) Pipeline-Based Approaches: Pipeline-based systems break the diagnosis process into sequential stages like similar to how clinicians approach diagnosis. Jundaeng et al. [4] implemented a precise periodontal pipeline: YOLOv8 Stage 1 for tooth segmentation (mAP50 = 0.92) feeds Stage 2 for CEJ and bone crest detection (mAP50 = 0.995, Sensitivity = 1.0). Pornprasertsuk-Damrongsri et al. [16] combined YOLOv5s for tooth detection with Attention U-Net for caries segmentation (Recall = 0.96, F1 = 0.85). A key limitation is error propagation where mistakes in early stages affect later stages. None of the reviewed pipelines address severity staging, and explainability appear only in one system [24] – which operates on periapical rather than panoramic, radiographs.

4) End-to-End Approaches: End-to-end models map radiographs to diagnostic outputs using deep architectures, primarily transformer-based models. Bhoopalan et al. [5] applied Vi-

sion Transformer (ViT) and Swin Transformer to 5,098 OPGs, achieving ViT accuracy = 96% without spatial localization or severity grading. Van Nistelrooij et al. [30] combined multiple datasets and transformer-based architectures to improve tooth-level analysis, although computational complexity and inference time remain challenges. The fundamental barrier is data: the field lacks large-scale, multi-disease, multi-center public datasets required for end-to-end learning.

The system-level taxonomy and capability gaps across different architectural paradigms are illustrated in Fig. 2

3.3 Task Hierarchy

Dental radiograph analysis follows a structured sequence of interdependent processing stages reflecting the actual cognitive process trained clinicians employ:

Early stages focus on anatomical structure identification, where systems perform well: tooth segmentation achieves pooled sensitivity = 0.92, specificity = 0.94 [28]; tooth enumeration reaches mAP50 = 0.978 [3]. Middle stages – disease detection and severity assessment – represent the performance bottleneck. Disease detection on OPG reaches only mAP50 $\approx$ 0.61 [3], and severity staging is attempted in only two of 18 C1 papers [15], [20]. Terminal stages – explainability and structured reporting – are most neglected: explainability appears in only 3 of 47 papers (6.4%) [10], [14], [24], and structured report generation is absent from all systems. This hierarchy shows that dental AI is a sequential reasoning problem, where failures in early stages can affect downstream diagnostic decisions. For systems targeting junior clinicians, the absence of visual explanations reduces its educational value.

3.4 System-Level Limitations

Five system-level limitations persist across the reviewed literature, reflecting field-wide methodological choices:

- **Task Fragmentation:** No panoramic OPG system simultaneously addresses caries detection, severity staging, periodontal quantification, and explainability. Chen et al. [24] achieves dual-disease detection with explainability – closest to unified – but only on periapical radiographs.
- **Severity Staging Deficit:** Only Asadi [20] and Bizjak and Robic [15] provide caries severity differentiation among 18 C1 papers. For periodontal disease, the 2018 AAP/EFP Stages I–IV [21] are not addressed by existing systems.
- **Limited Explainability:** Only 3 of 47 papers implement explainability (6.4%), despite a general consensus in the field [2], [3], [9], [35], [36]. The EU AI Act and FDA guidance require transparency for high-risk systems – requirements 43 of 47 systems do not satisfy.
- **Dataset Fragmentation:** Approximately 60% of studies use private, single-center datasets which lead to heterogeneity values $I^2 = 88$–96% [12], [27], [28]. Mureşanu et al. [7] documented 22% F1 degradation from internal to external data. There is currently no large-scale, multi-disease, multi-center, publicly available panoramic dataset with standardized annotations.
- **Clinical Workflow Disconnection:** With the exception of Schwendicke et al. [25] (cost-effectiveness analysis) and Lee et al. [26] (65-dentist usability study), no system evaluates AI within actual clinical workflows. Most systems produce outputs such as bounding boxes without structured reports or treatment recommendations..

4. Background

4.1 Panoramic Radiography in Clinical Dentistry

Orthopantomography (OPG) produces a two-dimensional representation of the curved dental arches on a flat image plane. Unlike periapical or bitewing radiographs that show specific regions, a single panoramic image captures all 32 permanent teeth, maxilla and mandible, and surrounding structures – making it one of the most informative radiographic methods in routine dental practice. For caries detection, panoramic radiographs are less sensitive than bitewing images for early proximal lesions [15], [16], [33]. For periodontal bone loss assessment, OPG is commonly used as a screening tool, as it allow visualization of the alveolar crest relative to the cemento-enamel junction (CEJ) across all teeth [2], [4], [27]. This difference in imaging affects how AI systems are designed for each task.

4.2 Deep Learning Architectures

The architectures reviewed can be divided into four major categories. First, CNNs with transfer learning, such as ResNet [32], DenseNet [17], [19], VGG [18], EfficientNet [24], Inception [34], [35] are mostly used for classification tasks, These models often use ImageNet pre-training because dental datasets are limited. Second, YOLO-family one-stage object detectors (YOLOv5–YOLOv9) [3], [7], [13], [15], [16] They are commonly used for detection tasks because they are fast and provide accurate bounding box localization. [6], [31], [37]–[39] are used for pixel-level segmentation, where skip connections help preserve spatial details. Fourth, transformer architectures – ViT [5], Swin Transformer [30], [40], RT-DETR [41] – represent newer approaches in this area, offering global attention but with higher computational cost.

4.3 Evaluation Metrics

A persistent challenge across the reviewed literature is metric inconsistency that prevents direct cross-study comparison [2], [6], [27], [28]. Object detection studies report mAP at IoU thresholds of 0.50 and 0.50:0.95. Segmentation studies report IoU and Dice coefficient. Classification studies report accuracy, AUC, sensitivity, specificity, and F1. Several studies [10], [24], [34] additionally report PPV and NPV – metrics with direct clinical relevance that standard accuracy obscures. The absence of a standardized dental AI evaluation protocol is identified as one of the field's most critical methodological gaps [1], [2], [27].

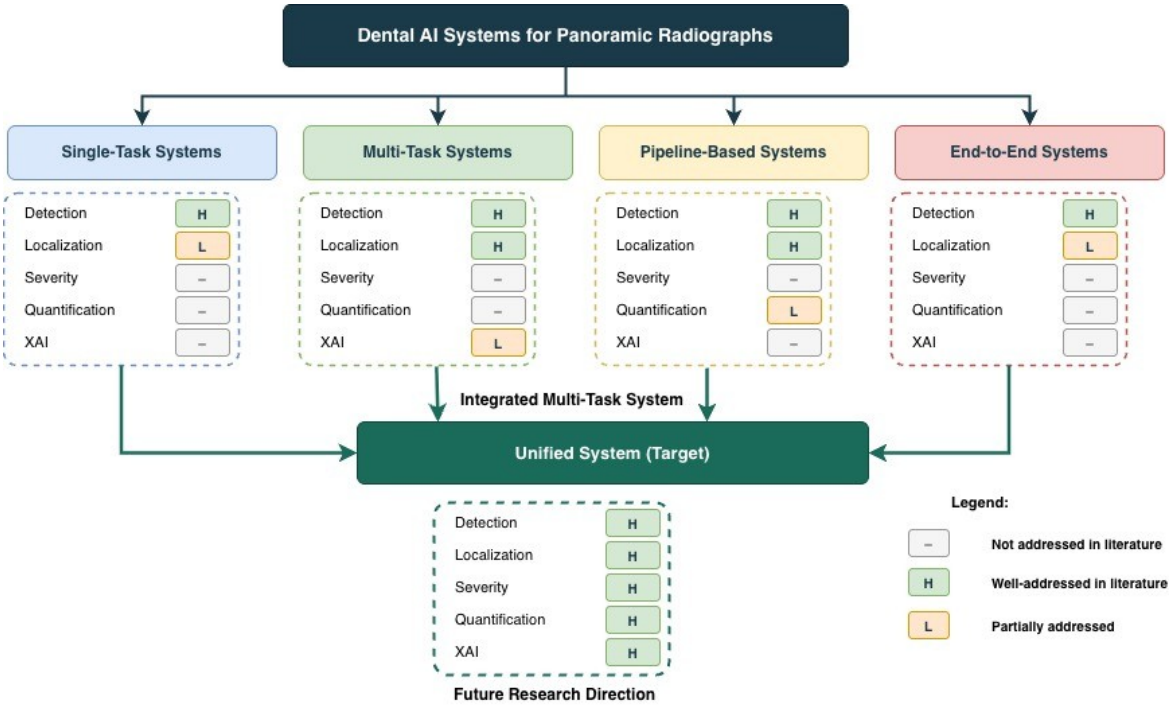


Fig. 2. System-level taxonomy of dental AI approaches for panoramic radiographs, illustrating four architectural paradigms (Single-Task, Multi-Task, Pipeline-Based, and End-to-End Systems) and their respective capability coverage across detection, localization, severity, quantification, and explainability (XAI) dimensions, alongside the proposed Unified System as a future research direction.

TABLE I
SYSTEM TAXONOMY — REPRESENTATIVE STUDIES BY DESIGN PATTERN AND CAPABILITY COVERAGE

System Type	Representative Studies	Detect	Localize	Severity	Quant.	XAI	Multi-Disease
Single-Task	[13], [14], [19], [31]–[34]	✓	Partial	×	×	×	×
Multi-Task	[3], [5], [7], [24], [30], [35]	✓	✓	×	×	Partial	Partial
Pipeline-Based	[4], [14]–[16]	✓	✓	×	Partial	×	×
End-to-End	[5], [30]	✓	Partial	×	×	×	Partial
Unified (Required)	*None in reviewed corpus*	✓	✓	✓	✓	✓	✓

✓ = fully addressed; × = not addressed; Partial = partially addressed.
Quant. = quantitative measurement; XAI = explainable AI outputs.

5. CARIES DETECTION AND SEVERITY CLASSIFICATION (C1)

Caries detection is the most extensively studied task in dental AI, accounting for 18 of the 47 reviewed papers. The Studies cover object detection, segmentation, classification, and systematic review methodologies across panoramic, bitewing, periapical, and intraoral RGB modalities.

5.1 YOLO-Based Detection on Panoramic Radiographs

Ayhan et al. [13] proposed a two-stage pipeline for detecting caries beneath fixed dental prostheses (FDPs) – a challenging clinical scenario where metal artifacts from crowns and bridges can obscure lesion visibility. YOLOv7 was used to detect FDP regions (mAP = 0.968, F1 = 0.956) in Stage 1; a CBAM-enhanced YOLOv7 applied to cropped ROIs achieved caries detection with mAP = 0.846 and F1 = 0.830 in Stage 2. The CBAM attention module improved detection of low-contrast lesion boundaries beneath metallic artifacts.

Bizjak and Robic [15] used a two-step YOLOv8 pipeline on the public DENTEX dataset, achieving tooth enumeration mAP@0.5 = 0.95 and lesion sensitivity = 0.70 (internal), 0.67 (external) – This is one of the few studies that includes external clinical validation, showing performance degradation from internal to external data. Asadi [20] used Faster R-CNN on the UFBA-UESC public dataset, achieving 94% overall accuracy and reported an important limitation: recall for mild caries = 0.64 versus severe caries = 0.69 – which is not ideal in practice, since early lesions are more important to detect. Pornprasertsuk-Damrongsri et al. [16] combined YOLOv5s for tooth detection with Attention U-Net for caries segmenta-

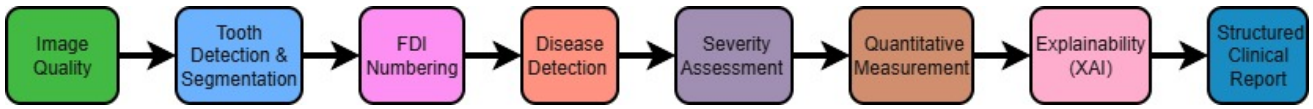


Fig. 3. Task hierarchy for panoramic dental radiograph analysis, showing stages from anatomical structure extraction to clinical report generation. Early stages, such as tooth segmentation and numbering, usually perform well. However, later stages like severity assessment, quantification, and explainability are less explored, which highlights current limitations.

tion on 500 panoramic radiographs with bitewing-confirmed ground truth, achieving Recall = 0.96, F1 = 0.85 with Bland-Altman agreement analysis.An example of multi-tooth carious lesion detection with severity differentiation in panoramic radiographs is shown in Fig. 4 .

5.2 U-Net Segmentation Approaches

Alharbi et al. [31] compared U-Net, U-Net++, and U-Net3+ on 1,500 panoramic images. U-Net3+ achieved the best test performance (IoU = 0.67, Dice = 0.60, Accuracy = 0.95), with its full-scale skip connections providing measurable advantage over standard U-Net (Dice = 0.58). The gap between training IoU (0.77) and test IoU (0.67) reveals overfitting from severe class imbalance – only 510 of 1,500 images contained visible cavities.

Lim et al. [38] proposed Faster R-CNN + U-Net (EfficientNet-B0 encoder) on 597 panoramic radiographs, achieving IoU = 0.608, Dice = 0.743, Recall = 0.731 on 33,752 augmented cropped tooth images, though without severity grading or explainability. Asci et al. [37] applied U-Net to 6,075 pediatric panoramic radiographs stratified across primary, mixed, and permanent dentition, achieving F1 = 0.8675, Sensitivity = 0.8269, Precision = 0.9123 – demonstrating U-Net's adaptability across diverse tooth morphologies, though AUC = 0.76 remains below the 90% clinical threshold. Vickram and Mathan Muthu [29] applied UNet variants on 500 dental X-ray images, achieving Accuracy = 98.03% and Dice = 96.77%, though on a notably small dataset with no clinical validation.

5.3 CNN Classification and Transfer Learning

Zancan et al. [34] applied InceptionResNet-v2 to 7,304 tooth-level images from 935 panoramic radiographs, achieving accuracy = 0.94 and F1 = 0.94 with five-fold cross-validation – one of the stronger validation approaches in C1. Kaushik and Sharma [17] applied DenseNet201 to 29,815 dental X-ray images, achieving 91% overall accuracy but a cavity-specific F1 of only 0.27 – a stark illustration of class imbalance consequences. Ozdemir et al. [19] applied DenseNet-121 to 14,498 cropped tooth images (2,303 caries / 12,195 non-caries), achieving accuracy = 93.17% and F1 = 87.49%, though without localization on full panoramic images. Khuwaja et al. [32] evaluated YOLOv8, ResNet-50, EfficientDet, and MobileNetV2 on 7,447 intraoral RGB smartphone images – targeting mobile diagnostics for resource-limited settings – with ResNet-50 achieving mAP@0.50 = 0.670.

5.4 Systematic Reviews and Meta-Analyses

Albano et al. [12] reviewed 20 studies via PRISMA/QUADAS-2, finding accuracy 73%–98% and AUC 0.84–0.98 across CNN-based models, with high heterogeneity and an explicit call for standardized datasets and multicenter validation. Luke and Rezallah [1] conducted a meta-analysis of 21 studies (17,190 total samples), reporting the best individual study achieving 98.8% accuracy but I^2 values of 88–96% indicating severe heterogeneity that undermines pooled estimates. Zanini et al. [6] reviewed 42 studies confirming that most works are single-task and lack caries severity differentiation as a systematic field-wide gap. Sastika and Marappan [36] reviewed accuracy ranges of 85–99% across multiple architectures, notably acknowledging Grad-CAM as a requirement in surveyed studies.

Chen et al. [14] introduced the CariesXrays dataset (6,000 hospital-grade panoramic X-rays, 13,783 expert-verified bounding boxes) alongside the FPCL architecture – achieving AP = 48.2%, AP50 = 84.1%, outperforming YOLOv8 (AP50 = 80.7%). CariesXrays is the most significant public benchmark contribution in the C1 category.

5.5 Bitewing Caries and Cost-Effectiveness

Perez de Frutos et al. [33] evaluated RetinaNet, YOLOv5-M, and EfficientDet on 13,887 bitewing X-rays from the HUNT4 Oral Health Study, annotated by six dentists with a 197-image consensus test set. YOLOv5-M achieved mAP = 0.647, F1 = 0.548 – matching or surpassing individual expert annotators, providing strong evidence that deep learning performs at human expert level on proximal caries detection. Schwendicke et al. [25] combined U-Net performance evaluation (accuracy = 0.80, sensitivity = 0.75 vs. dentist sensitivity = 0.36) with a Markov lifetime cost-effectiveness simulation on 3,686 bitewing radiographs, projecting increased tooth retention (64 vs. 62 years) and reduced cost per tooth (€298 vs. €322), yielding ICER = −13.9 €/year – establishing AI caries detection as a cost-dominant strategy.

6. Periodontal Disease and Alveolar Bone Loss Detection (C2)

Periodontal disease detection from radiographs requires quantifying the spatial relationship between anatomical landmarks – specifically the distance between the cemento-enamel junction (CEJ) and the alveolar bone crest – across all teeth in the panoramic image. This requires precise spatial localization and continuous measurement rather than binary classification. Fifteen papers in the C2 category indicate that this area is still developing but progressing rapidly.

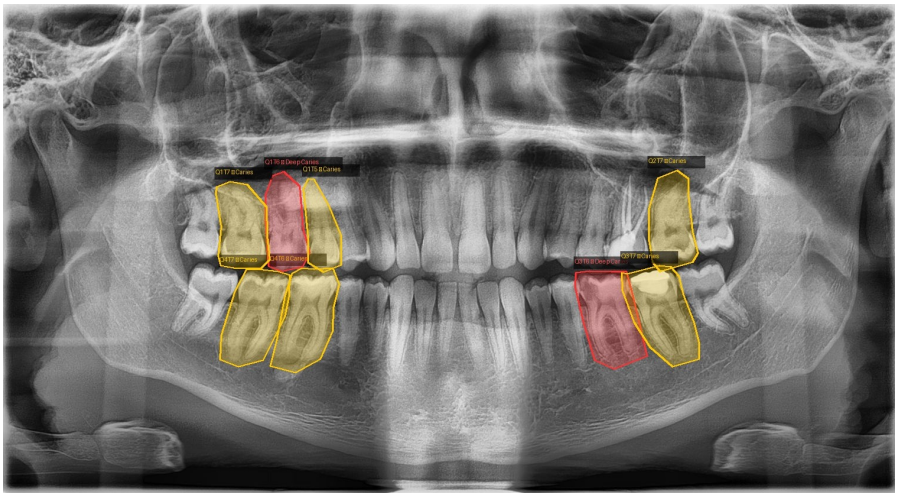


Fig. 4. Example panoramic radiograph with annotated carious lesions, illustrating multi-instance detection and severity differentiation (caries vs. deep caries) across multiple teeth.

6.1 Clinical Foundation

Chmielewski et al. [21] provide essential clinical framing: the 2018 AAP/EFP classification defines periodontitis stages I–IV and grades A–C based on bone loss percentage, probing depth, tooth loss, and risk factors. Across 11 clinical studies covering 459 cases and 852 examiners, staging accuracy peaked at 90% but inter-examiner agreement remained variable – the subjectivity problem that AI systems can address, but only if they produce quantitative, staged outputs aligned with the 2018 criteria rather than binary bone loss labels. Fig. 5 illustrates the spatial nature of periodontal bone loss assessment, requiring accurate identification of anatomical landmarks such as the cemento-enamel junction and alveolar bone crest.

6.2 Landmark Detection and Bone Loss Quantification

Jundaeng et al. [4] presented the most technically complete periodontal AI system reviewed. Their dual YOLOv8 pipeline on 2,000 panoramic radiographs achieved tooth segmentation (Accuracy = 0.97, mAP50 = 0.92) and CEJ/bone level detection (Accuracy = 0.98, Sensitivity = 1.0, Specificity = 0.98, F1 = 0.90, mAP50 = 0.995). Sensitivity = 1.0 for bone level detection is clinically important – missing true bone loss is more harmful than a false positive in a screening context. Limitations include single-center data, high-quality images only, and no caries integration or explainability.

Iacob et al. [27] synthesized evidence from 20 diagnostic accuracy studies covering 141,166 radiographs through PRISMA/QUADAS-2, finding that U-Net, YOLO, ResNet, ConvNeXt, and DeNTNet have all been applied but high heterogeneity prevents reliable pooling. Khubrani et al. [2] meta-analyzed 30 studies using APPRAISE-AI, reporting pooled sensitivity = 87%, specificity = 76%, accuracy = 84% – useful but insufficient for autonomous clinical use.

Mathur et al. [18] applied VGG-16 to 1,874 panoramic radiographs for binary bone loss classification (accuracy = 89.27%, sensitivity = 83.27%) – the most common C2 approach but also the most clinically limited, providing no staging information. Cerda Mardini et al. [42] attempted quantitative staging on 500 panoramic radiographs, reporting F1 = 0.29 for light-moderate bone loss and F1 = 0 for severe bone loss – honestly documenting the difficulty of quantitative staging and the model's failure on the most clinically critical class. Kong et al. [39] integrated SRGAN super-resolution preprocessing with U-Net for bone loss segmentation (Dice = 0.91, IoU = 84.9%) – the strongest segmentation metrics in C2 – demonstrating that preprocessing quality directly impacts segmentation accuracy, though the dataset of 698 images is small and no disease staging is provided.

Wimalasiri et al. [22] applied YOLOv8x and Keypoint R-CNN to 1,000 intraoral periapical X-rays, achieving ICC = 0.80 for severity and 87% for bone loss pattern classification (horizontal vs. vertical) – the only study addressing pattern classification, which has direct prognostic implications for surgical planning. Tian et al. [23] developed the PANPOL mathematical model validated on 290 panoramic patients, achieving ICC = 0.83, though this is a rule-based approach rather than a deep learning system.

6.3 Periodontitis Classification

Bilal et al. [10] proposed PDCNET – a custom five-block CNN with Grad-CAM explainability – compared against EfficientNet-B0, DenseNet-201, VGG-16, VGG-19, Inception-V3, and MobileNet on a Kaggle dataset augmented with SMOTE-Tomek, achieving accuracy = 98.39%, AUC = 99.79%, F1 = 98.31%. These are the highest classification metrics in C2, though the 50+50 original sample size raises serious concerns about statistical meaningfulness. The paper is valuable for demonstrating Grad-CAM feasibility in periodontal AI. Ferrara et al. [9] reviewed 13 periodontal deep learning studies via PRISMA/QUADAS-2, reporting accuracy ranges of 73%–98.6% and noting the complete absence of

TABLE II
SUMMARY OF REVIEWED STUDIES — CARIES DETECTION AND SEVERITY CLASSIFICATION (C1)

Ref	Model	Dataset Size	Best Metric	XAI	Key Gap
[13]	YOLOv7 + CBAM	1,004 OPG	mAP=0.846, F1=0.830	None	Single-center, no XAI
[32]	YOLOv8, ResNet-50, EfficientDet, MobileNetV2	7,447 RGB	mAP@0.50=0.670	None	RGB only, no radiographs
[31]	U-Net, U-Net++, U-Net3+	1,500 OPG	IoU=0.67, Dice=0.60	None	Class imbalance, no severity
[37]	U-Net	6,075 OPG (pediatric)	F1=0.8675, AUC=0.76	None	Below 90% clinical threshold
[12]	Systematic Review (CNNs)	20 studies	Acc up to 0.98, AUC 0.98	None	High heterogeneity
[17]	DenseNet201	29,815 X-rays	Acc=91%, Cavity F1=0.27	None	Severe class imbalance
[14]	FPCL (Faster R-CNN + D2D-FPN + Contrastive)	6,000 OPG	AP=48.2%, AP50=84.1%	Attn maps	No severity/periodontal
[33]	YOLOv5-M, RetinaNet, EfficientDet	13,887 bitewing	mAP=0.647, F1=0.548	None	Bitewing only, no periodontal
[36]	Review: U-Net, ResNet, YOLOv7/v8, ensemble	Multiple studies	Acc 85–99%, F1 87–94%	Discussed	No original experiment
[25]	U-Net	3,686 bitewing	Acc=0.80, ICER=−€13.9/yr	None	Single-center, German context
[1]	Meta-analysis (CNNs)	21 studies, 17,190 images	Best Acc=98.8%	None	I^2=88–96% heterogeneity
[29]	UNet, UNet+, UNet++	500 images	Acc=98.03%, Dice=96.77%	None	Small dataset, no clinical val.
[34]	InceptionResNet-v2	935 OPG / 7,304 teeth	Acc=0.94, F1=0.94	None	No localization, no periodontal
[38]	Faster R-CNN + U-Net (EfficientNet-B0)	597 OPG	IoU=0.608, Dice=0.743	None	No severity, no periodontal
[6]	Systematic Review (U-Net, Mask R-CNN, ResNet)	42 studies	Various (review)	None	No severity differentiation
[15]	YOLOv8 two-step pipeline	1,005 OPG + 85 external	mAP@0.5=0.95, Sens=0.70	None	Caries only, no severity
[20]	Faster R-CNN	1,000 OPG	Acc=94%, Mild recall=0.64	None	Weak mild caries detection
[16]	YOLOv5s + Attention U-Net	500 OPG, 14,997 teeth	Recall=0.96, F1=0.85	None	Single-disease, no periodontal
[19]	DenseNet-121	14,498 tooth images	Acc=93.17%, F1=87.49%	None	No localization, no severity

explainable AI across reviewed systems. Lee et al. [26] evaluated AI alveolar bone loss measurement usability through a survey of 65 dentists on 550 bitewing images – one of the only clinical usability studies in the entire reviewed corpus.

Kim et al. [43] proposed an unsupervised few-shot learning architecture combining UNet, CVAE, and clustering methods (k-means/DBSCAN/GMM) on 1,372 panoramic radiographs, achieving a 14% improvement over supervised ViT, demonstrating the potential of label-efficient learning in data-scarce dental AI settings. Karamuftuoğlu et al. [41] benchmarked YOLOv12x, RT-DETR-L, and RT-DETR-X on a 387-image pediatric panoramic dataset for furcation lesion detection, achieving mAP@0.5 = 0.434 – reflecting both the difficulty of the task and the limitation of small single-center datasets.

7. MULTI-TASK AND END-TO-END OPG DIAGNOSIS (C3)

Seven papers address multiple diagnostic tasks within a single pipeline applied to panoramic radiographs – the architectural direction most aligned with real clinical utility.

Chen et al. [24] present the most clinically complete dual-disease system reviewed, combining YOLOv7 for tooth localization with EfficientNet-B0 for simultaneous multi-label classification of both periodontitis and dental caries on periapical X-rays, with Grad-CAM visualization. EfficientNet-B0 achieved ROC AUC = 98.67% for periodontitis and 98.31% for dental caries with 10-fold cross-validation. The critical limitation is modality: periapical, not panoramic.

Nassiri and Akhloufi [3] systematically compared YOLO v5–v9 on DENTEX across quadrant segmentation, tooth enumeration, and disease detection. YOLOv9e-seg achieved mAP50 = 0.978 for tooth enumeration and 0.959 for segmentation, but pathology detection peaked at mAP50 = 0.61 – confirming that structural tasks are reliably solvable while pathology detection remains the harder problem.

Van Nistelrooij et al. [30] combined OdontoAI and DENTEX datasets with Mask DINO (ViT backbone) and Swin-B Transformer, achieving FROC = 0.988 for tooth segmentation and AUC = 0.996 for impacted teeth – the highest-performing tooth-level assessment reviewed. The 40-second inference time per image is the primary barrier to clinical deployment. Gamal et al. [35] demonstrated that a Weighted Average Ensemble of

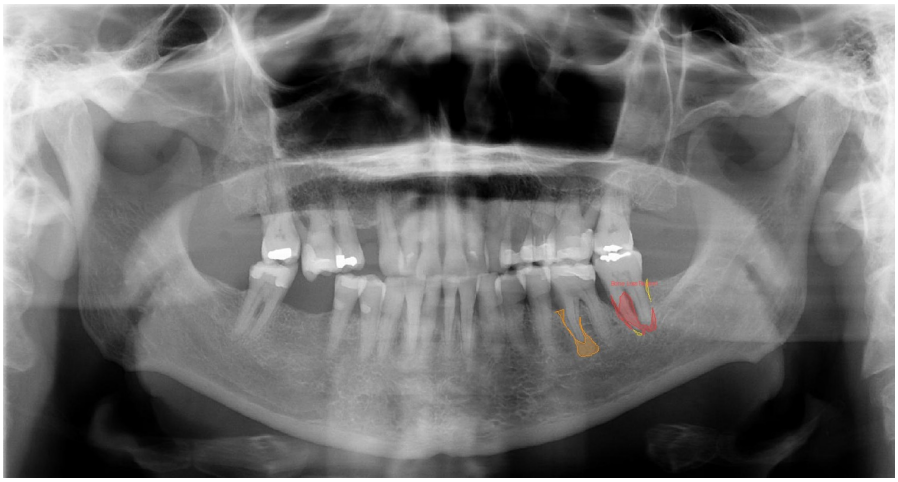


Fig. 5. Example panoramic radiograph illustrating periodontal bone loss regions. Accurate assessment requires spatial measurement between the cemento-enamel junction and alveolar bone crest.

CNN architectures (Custom CNN, ResNet-50, InceptionV3, DenseNet121, VGG19) achieves accuracy = 92.93% on 1,814 cropped tooth images, with ESRGAN super-resolution preprocessing contributing approximately 3% accuracy improvement.

Mureşanu et al. [7] has given the most methodologically reliable external validation in C3. When YOLOv8 was tested on a Romanian clinical dataset,internal F1 = 0.60 reduced to F1 = 0.47 on external data,showing a 22% performance drop . This clearly justifies the gap between controlled training and real multi-center clinical data. Bhoopalan et al. [5] applied Vision Transformer(ViT) and Swin Transformer to 5,098 OPG images, achieving a ViT accuracy of 96%, with precision of 95.8% and recall of 96.2% .These results are strong enough for image-level classification, even though localization or segmentation were not included in the approach. Zhou et al. [11] conducted a comprehensive systematic review of datasets, methodologies, and emerging challenges in dental image analysis, highlighting the need for standardization in the field.

8. Explainable AI in Dental Diagnostics (C4)

Of the 47 papers reviewed, only three implement any form of explainability output.Chen et al. [24] use Grad-CAM for dual-disease classification on periapical X-rays; Bilal et al. [10] apply Grad-CAM to periodontal disease classification; and Chen et al. [14] show attention visualization in feature maps for caries detection. This constitutes **6.4%** of the reviewed corpus – directly quantifying the explainability gap in dental AI.

Gradient-weighted Class Activation Mapping (Grad-CAM) produces a coarse localization map indicating which image regions most influenced a CNN's prediction. In the dental diagnostics, Grad-CAM serves three clinical functions: (i) it allows clinicians to confirm that the model attends to anatomically plausible regions; (ii) it gives junior clinicians a visual reference for learning radiographic interpretation; and (iii) it generates an audit trail relevant to the EU AI Act and FDA guidance on AI/ML-based medical devices – both of which classify AI diagnostic tools as high-risk systems requiring transparency and human oversight [24].

There is a broad agreement in the literature that explainability is required for clinical adoption [3], [9], [35], [36]. However, the adoption rate remains at 6.4% . The gap becomes even more critical for junior clinicians and dental students, who require not only prediction but also visual explanations to help them in learning. None of the reviewed dental AI study implements methods like SHAP, LIME,or transformer attention visualization, which represents a clear gap, given that transformer attention maps come at no extra computational cost [5].

9. Comparative Analysis of Representative Methods

10. Datasets, Benchmarks, and Clinical Validation (C5)

Dataset availability and quality are the most consistently cited limitation across all 47 reviewed papers. The key public benchmarks in the field are as follows:

- **DENTEX**: 1,005 panoramic radiographs with quadrant, tooth, and pathology annotations from three institutions [3], [15], [30].
- **CariesXrays**: 6,000 hospital-grade panoramic images with 13,783 caries bounding boxes [14].
- **Tufts Dental Database**: Approximately 1,000 panoramic radiographs with tooth segmentation masks [40], [42]–[44].
- **HUNT4 Oral Health Study**: 13,887 bitewing X-rays with six-dentist annotations and a 197-image consensus set [33].
- **UFBA-UESC**: 4,000 panoramic images [20].

Despite these contributions, roughly 60% of reviewed primary studies rely exclusively on private, single-center institutional datasets. Systematic reviews [1], [2], [12], [27], [28]

TABLE III
SUMMARY OF REVIEWED STUDIES — PERIODONTAL DISEASE AND BONE LOSS (C2)

Ref	Model	Dataset Size	Best Metric	XAI	Key Gap
[4]	Dual YOLOv8 (teeth + CEJ/bone)	2,000 OPG	mAP50=0.995, Sens=1.0	None	Single-center, no caries
[21]	Clinical review (no ML)	459 cases, 852 examiners	Stage acc=90%	None	No AI system proposed
[27]	Systematic Review (CNN, YOLO, U-Net)	141,166 radiographs	Various (review)	None	High heterogeneity
[18]	VGG-16 (transfer learning)	1,874 OPG	Acc=89.27%	None	Binary only, no staging
[2]	Meta-analysis (CNN, YOLO, U-Net, SVM)	30 studies	Sens=87%, Spec=76%	None	Pooled heterogeneity
[23]	PAN-POL mathematical model	290 PAN patients	ICC=0.83	None	Not deep learning, manual
[26]	Faster R-CNN + DeepLab v3+	550 bitewing	Acc=82–87% (severe)	None	Bitewing only, no OPG
[42]	Hybrid DCNN (Xception) + rule-based PBL	500 OPG	F1=0.29 (lt-mod), F1=0 (severe)	None	Fails on severe PBL
[22]	YOLOv8x + Keypoint R-CNN	1,000 IOPA	ICC=0.80, Pattern Acc=87%	None	Periapical only, no OPG
[39]	SRGAN + U-Net	698 OPG	Dice=0.91, IoU=84.9%	None	No severity, small dataset
[9]	Systematic Review (CNNs, hybrid)	13 studies	Acc 73–98.6%	None	No XAI in any reviewed study
[10]	PDCNET (custom CNN) + Grad-CAM	50+50 orig (SMOTE aug.)	Acc=98.39%, AUC=99.79%	Grad-CAM	Dataset too small
[41]	YOLOv12x, RT-DETR-L, RT-DETR-X	387 OPG (pediatric)	mAP@0.5=0.434	None	Small dataset, modest mAP
[43]	UNet + CVAE + k-means/DBSCAN/GMM	1,372 OPG	+14% over supervised ViT	None	No bounding boxes

consistently report I^2 heterogeneity values of 88–96% as the main barrier to reliable evidence synthesis. The performance degradation from internal to external validation quantified by Mureşanu et al. [7] – F1 dropping from 0.60 to 0.47 – directly measures the real cost of single-center training.

Waqas et al. [45] addressed data scarcity through GAN-based synthetic OPG generation (DCGAN, WGAN) on 4,777 real images, producing synthetic panoramic radiographs with expert realism assessed above 50% relative to real images – a promising augmentation approach, though downstream disease detection improvement was not demonstrated. Ameli et al. [46] applied YOLOv8 to 1,000 OPGs for automated quality assessment (contrast/density accuracy = 97.9%, artifact accuracy = 87.2%, overall quality = 79.3%) – establishing automated quality filtering as a viable preprocessing gate for diagnostic AI pipelines. Delamare et al. [8] documented heterogeneous quality criteria across 41 panoramic ML studies, calling for standardized quality protocols as a prerequisite for reproducible AI development.

11. TOOTH SEGMENTATION AND NUMBERING (C6)

Four papers address tooth detection, segmentation, and numbering – enabling upstream tasks that are prerequisites for tooth-specific disease attribution. Accurate FDI-standard tooth numbering is what allows a system to report that caries is located in tooth 26 rather than somewhere unspecied in the panoramic image.

Bonfanti-Gris et al. [28] synthesized 20 tooth detection and segmentation studies in a PRISMA/QUADAS-2 meta-analysis, reporting pooled sensitivity = 0.92 and specificity = 0.94 with DOR = 186 – strong aggregate evidence, though $I^2 \approx 96\%$ confirms severe heterogeneity. Zhang et al. [40] proposed a multi-stage system (MobileNetV3 + SADP-Net Mask R-CNN + Swin Transformer + YOLOv5) across three public datasets, achieving AP50 = 97.3% on Tufts and 98.1% on O2PR for tooth instance segmentation. Yaswanth et al. [44] applied Attention-integrated U-Net++ to the Tufts dataset, reaching mAP = 96.4% and Dice = 90%. Adnan and Umer [47] released the Orthopantomogram Teeth Segmentation and Numbering Dataset, consisting of 250 expert-annotated OPGs – exclusively highlighting the need for including disease annotations.

12. CROSS-STUDY TRENDS AND OBSERVATIONS

This section summarizes the commonly occurring patterns identified across the 47 reviewed studies. From the global comparison in Table IV, different trends are observed in terms of model design, dataset characteristics, task complexity, and deployment constraints.

12.1 Model Architecture Trends

1) Transformer-Based vs. CNN-Based Architectures: Transformer-based models, such as Vision Transformer(ViT)

TABLE IV
COMPARISON OF REPRESENTATIVE DEEP LEARNING METHODS FOR DENTAL RADIOGRAPH ANALYSIS. PERFORMANCE METRICS ARE REPORTED AS PROVIDED IN THE RESPECTIVE STUDIES AND ARE NOT DIRECTLY COMPARABLE DUE TO VARIATIONS IN TASKS, DATASETS, AND EVALUATION PROTOCOLS.

Ref	Task	Scope	Dataset (Size)	Modality	Model	Best Metric	XAI	Clinical Val.	Key Limitation
[13]	Caries Detection	Single-task	Private OPG (1,004)	OPG	YOLOv7 + CBAM	mAP=0.846	No	Internal	No external validation
[4]	Bone Loss + Periodontal	Multi-task	Private OPG (2,000)	OPG	YOLOv8	mAP50=0.995	No	Internal	Single-center dataset
[24]	Multi-disease Detection	Multi-task	Periapical (11k)	X-ray	YOLOv7 + EfficientNet	AUC≈98%	Yes	Internal	Not panoramic
[31]	Caries Segmentation	Single-task	DNS (1,500)	OPG	U-Net3+	Dice=0.60	No	Internal	Small dataset
[37]	Caries Segmentation	Single-task	Pediatric OPG (6,075)	OPG	U-Net	F1=0.867	No	Internal	No severity staging
[35]	Multi-disease Classification	Multi-task	Roboflow (1,075)	OPG	CNN Ensemble	Acc=92.9%	No	Internal	No explainability
[14]	Caries Detection	Single-task	CariesXrays (6,000)	OPG	Faster R-CNN	AP50=84.1%	Partial	Internal	Single-disease
[33]	Caries Detection	Single-task	HUNT4 (13k)	Bitewing	YOLOv5	mAP=0.647	No	External	Not panoramic
[3]	Multi-task OPG Analysis	Multi-task	DENTEX (∼1,000)	OPG	YOLOv9-seg	mAP=0.978	No	Internal	Limited pathology
[30]	Tooth + Diagnosis	Multi-task	OdontoAI (1,661) + DENTEX (682)	OPG	Mask DINO + Swin	mAP=0.848	No	Internal	Slow pipeline
[34]	Tooth + Caries	Multi-task	935 PAN + 18k teeth	OPG	InceptionResNet	Acc=0.998	No	Internal	No localization
[38]	Caries Segmentation	Single-task	597 PAN	OPG	Faster R-CNN + U-Net	Dice=0.743	No	Internal	Single-disease
[15]	Caries Detection	Single-task	DENTEX + external	OPG	YOLOv8	mAP=0.95	No	External	No segmentation
[20]	Caries Severity	Single-task	1,000 PAN	OPG	Faster R-CNN	Acc=94%	No	Internal	Weak mild detection
[16]	Caries Detection	Multi-stage	500 PAN	OPG	YOLOv5 + U-Net	Recall=0.96	No	Internal	Single-disease
[18]	Bone Loss Classification	Single-task	1,874 OPG	OPG	VGG-16	Acc=89.27%	No	Internal	No localization
[26]	Bone Loss Detection	Single-task	Bitewing (550)	Bitewing	Faster R-CNN + DeepLab	Acc≈85%	No	Internal	Limited modality
[22]	Bone Loss Severity	Single-task	DenPAR (1,000)	IOPA	YOLOv8 + KRCNN	ICC=0.80	No	Internal	Not panoramic
[39]	Bone Loss Segmentation	Single-task	CBNUH (100) + Kaggle (∼600)	OPG	SRGAN + U-Net	Dice=0.91	No	Internal	No detection
[7]	Multi-disease Detection	Single-task	Multi-center panoramic	OPG	YOLOv8	F1=0.60→0.47	No	External	Poor generalization
[40]	Tooth Segmentation	Multi-stage	Tufts + O2PR	OPG	Mask R-CNN + YOLO	mAP=77.4	No	Internal	No disease labels
[28]	Tooth Segmentation Review	Review	20 studies	Multi	CNNs	Sens=0.92	No	Review	High heterogeneity
[2]	Bone Loss Review	Review	30 studies	Multi	CNNs	Acc=84%	No	Review	No standardization
[11]	Deep Learning Survey	Review	260 studies	Multi	Various	—	No	Review	No unified system
[10]	Periodontal Classification	Single-task	Kaggle (∼1,000)	X-ray	PDCNET	Acc=98.39%	Yes	Internal	No localization

and Swin Transformer can capture broad contextual information, which is particularly used for classification and multi-label diagnosis tasks [30]. However CNN-based architectures dominate the most of reviewed studies, especially in detection and pipeline-based systems [17]. Their efficiency, lower computational requirements, and their ability to perform well on moderately sized dental datasets make them more suitable for current applications.

Transformer-based models usually require large annotated datasets and have higher inference latency, which limits their practical use in clinical environments [30]. Given the current limitations in data availability, CNN-based approaches remain more suitable for real-world deployment.

2) Dominance of YOLO in Detection Tasks: YOLO-based architectures are most commonly used detection models in the reviewed set. As seen in the Table IV ,different YOLO variants appear across caries localization, tooth detection, and bone loss identification studies [3]. Their ability to deliver accurate results in a single inference pass with a good real-time performance, makes them most suitable for clinical use.

3) Ensemble Methods: Ensemble methods can improve classification robustness by combining multiple models, but introduce computational complexity and reduce inference – these tradeoffs limit their application in time-sensitive clinical settings [35].

12.2 Data and Generalization

Large datasets do not guarantee generalization . Class imbalance can undermine task-specific performance even with

the large datasets [17], and models trained exclusively on single-center data reliably degrade when evaluated externally.

This generalization gap is a structural feature of the field, corroborated by systematic reviews that attribute high inter-study heterogeneity to differences in imaging protocols, annotation standards, and architectures [2].Dataset diversity, multicenter representation, and standardized evaluation protocols are essential conditions –not optional refinements – for clinically reliable systems.

12.3 Task Complexity and Learning Paradigm

1) Structural Tasks vs. Disease Detection: The fundamental asymmetry in task difficulty runs through reviewed literature.Structural tasks – tooth segmentation and enumeration – are low-variance problems that achieve consistently high performance across architectures [28].Early-stage caries detection involves subtle and variable visual features, and performance drops accordingly.Even systems which perform well on structural tasks show lower accuracy when the same framework is used for pathological detection [3].

2) Multi-Task Learning vs. Single-Task Models: Single-task approaches dominate the reviewed studies, with multi-task approaches remain uncommon. There is a significant gap between current research practice and clinical needs: meaningful dental diagnosis requires evaluating multiple conditions at the same time. The limited number of multi-task systems that exists are often restricted in modality or scope [24].

12.4 Deployment and Evaluation Challenges

1) Explainable AI: Explainability is present in only about 6.4%of reviewed studies, leaving the most of systems as uninterpretable black-box models [24]. This lack of interpretability reduces clinical trust and makes regulatory approval difficult. For clinical deployment, explainability is essential rather than optional feature.

2) Lack of Standardized Evaluation Metrics: Metrics: Metric heterogeneity further sums up this problem. Studies report such as mAP, Dice, IoU, accuracy, and F1 using different thresholds, which makes direct comparisons across models unreliable.

Systematic reviews confirm that this variability is a major cause of differences between the studies [28]. Standardized evaluation protocols are necessary for meaningful progress.

12.5 Final Observation

CNN-based and YOLO-driven approaches currently define dental AI systems, while transformer-based models emerging as reliable but resource-intensive alternatives. Structural tasks usually achieve high reliability; disease detection, especially in early stages, along with quantitative analysis, remains significantly more challenging.Across all categories, there exists a clear disconnect that separates isolated high-performance results and fully integrated clinical systems.

13. Critical Analysis of Existing Methods

13.1 Metrics and Evaluation Limitations

The lack of standardized evaluation practices remains a persistent structural issue. Studies report metrics such as mAP, Dice coefficient, accuracy, AUC, and F1-score , which are found to be using inconsistent thresholds and experimental setups.This makes direct comparison among the different methods difficult and and makes it harder to find out which approaches are truly superior [2], [27], [28].

Many studies produce strong results under controlled conditions, but these may not hold under standardized benchmarking.Only a small number of works use evaluation settings that are comparable among the different studies.Without unified benchmarks, reproducibility and reliability continue to remain constrained.

13.2 Dataset Bias and Generalization Issues

The reliance on single-center dataset is a major constraint affecting generalizability. A substantial portion of the reviewed studies use data from a single institution or imaging setup, which does not capture variability in populations, devices, and acquisition conditions [1], [12].

Models trained in this manner often achieve strong performance on internal data but tend to perform worse when evaluated on external datasets, as reported across multiple studies [7].

Publicly available datasets remain limited in scale and diversity.DENTEX and CariesXrays provide useful baselines but do not adequately represent multi-disease, multi-center clinical variability [3], [14].

Without diverse, standardized, multi-institutional datasets and consistent validation protocols,the generalizability of current systems – and therefore their reliability in practice – remains questionable.

13.3 Clinical Integration Gap

Strong controlled-setting performance has not translated into clinical deployment.Most systems generate intermediate outputs – bounding boxes, segmentation masks, or classification labels – without connection to real-time decision support.

Clinician involvement in model development and evaluation is limited, and only a small number of studies assess real-world impact on diagnostic accuracy, treatment planning, or workflow efficiency [25], [26].

The limited adoption of explainable AI deepens this disconnect: operating as black-box models,most systems cannot satisfy the interpretability requirements of clinical practice.

13.4 Fragmentation of Existing Approaches

As seen in Table I, most reviewed studies address detection, segmentation, or classification in isolation. Real-world dental diagnosis requires simultaneous evaluation of caries detection, periodontal assessment, severity staging, and quantitative measurement.No reviewed system operating on panoramic radiographs jointly addresses all of these within a unified framework . This is a field-level structural limitation rather

than a failure of individual models. Resolving it requires a shift from task-specific optimization toward system-level design, where multiple diagnostic components are jointly modeled and aligned with actual clinical requirements.

14. Cross-Cutting Gap Analysis

Five critical interconnected gaps emerge from synthesizing all 47 reviewed papers, each consistently limiting clinical translational potential.

14.1 Gap 1: Single-Disease Focus

Single-disease training is the dominant across all categories.Among the 47 papers reviewed, only Chen et al. [24] simultaneously detect both caries and periodontal disease –and exclusively on periapical, not panoramic, radiographs. Particularly for panoramic OPGs, none of the reviewed study provides a single pipeline that does caries detection, severity staging, and periodontal bone loss quantification at the same time. This exists as the major architectural gap when compared to the real clinical scenarios, where patients often present with multiple conditions that need to be assessed simultaneously within in a single scan.

14.2 Gap 2: Absence of Automated Severity Staging

Among 18 C1 papers, only Asadi [20] and Bizjak and Robic [15] consider severity beyond simple binary caries detection. In the C2 category, only Wimalasiri et al. [22] examine bone loss severity patterns. The 2018 AAP/EFP classification [21] defines four periodontitis stages, none of which is addressed by any panoramic OPG system in the reviewed set. Severity staging plays a major role in determining treatment planning, referral decisions, and prognosis. A system that detects caries without assigning severity offers limited practical value.

14.3 Gap 3: Explainability at 6.4%

Only 3 out of 47 papers implement any form of XAI output. The broad agreement that explainability is essential for clinical adoption [3], [9], [35], [36] is not reflected in actual implementation. The EU AI Act classifies AI diagnostic tools as high-risk systems that require transparency and human oversight – requirements that most of the current dental AI systems do not satisfy. For junior clinicians, explainability is essential, as it allows AI-assisted diagnosis to be understood rather than treated as a black-box output.

14.4 Gap 4: Dataset Fragmentation

Around 60% of the reviewed studies rely on private, single-center datasets. Systematic reviews [1], [2], [12], [27], [28] consistently report heterogeneity levels of $I^2 = 88$–96% as the main barrier to evidence synthesis. The impact of this fragmentation has been highlighted by Mureşanu et al. [7], where F1 reduces by 22% when moving from internal to external validation. The field requires a single large-scale, multi-disease, multi-center, publicly available panoramic OPG dataset with standardized caries and periodontal annotations. Nothing in the reviewed literature currently provides this.

14.5 Gap 5: Clinical Workflow Integration

Excluding Schwendicke et al. [25] and Lee et al. [26], no reviewed study evaluates AI within an actual clinical workflow or measures clinician acceptance and diagnostic impact. Most systems produce bounding boxes or segmentation masks without generating clinical reports, severity summaries, or treatment recommendations. The economic analysis by Schwendicke et al. [25] demonstrating ICER = -13.9 €/year for AI-assisted caries detection, is an isolated example of the real-world impact analysis the field urgently needs at scale.

15. Research Roadmap Toward Unified and Clinically Deployable Dental AI Systems

Building on the limitations identified in Section 13, this section outlines a structuredprogression toward unified, clinically deployable dental AI.

15.1 Short-Term Goals (1–2 Years)

Public Multi-Disease Panoramic Dataset: The most immediate bottleneck is the absence of a large-scale, standardized panoramic dataset with multi-disease annotations. A consortium-driven effort is required to develop a dataset containing at least 5,000 panoramic radiographs with consistent per-tooth annotations for caries severity, periodontal bone loss , and tooth numbering FDI-standard tooth numbering.

CariesXrays [14] and DENTEX [3] offer partial coverage but lack comprehensive multi-disease labelling; extending and integrating these represents a practical short-term direction.

Explainability as a Standard Component: Explainability must also be treated as a standard component rather than an optional addition – via Grad-CAM or attention visualization, with validation against expert annotations to confirm anatomical relevance [10], [17].

15.2 Mid-Term Goals (3–5 Years)

Unified Multi-Disease System for Panoramic Radiographs: The most consequential research direction is a unified multi-disease system capable of simultaneously detecting caries,staging severity, quantifying periodontal bone loss, and supporting multi-condition diagnosis from a single panoramic radiograph. Multi-head architectures with shared feature representations and task-specific outputs are the likely architectural basis.Individual components exist in the literature [3], [4], [7], [24];the open problem is integrating them into a coherent, clinically usable framework.

Clinically Aligned Systems with Standardized Diagnostic Frameworks: Future systems should generate outputs that aligned with established clinical standards, such as ICCMS for caries and the 2018 AAP/EFP classification for periodontal disease [21].Aligning model outputs with these frameworks will help in generating clinically interpretable decisions.

15.3 Long-Term Goals (Clinical Deployment)

Prospective Clinical Validation and Deployment: Moving beyond retrospective evaluation requires prospective clinical validation, including randomized studies that assess diagnostic

accuracy, clinician efficiency, and patient outcomes, both with and without AI assistance,under standardized conditions.This is required to ensure reliable comparisons across different systems.

The cost-effectiveness framework proposed by Schwendicke et al. [25] can be extended to multi-disease systems using both bitewing and panoramic radiographs. Such economic validation is not only essential for regulatory approval but also for establishing clinical trust that enables adoption of AI systems in clinical settings.

Final Perspective: *Overall, this roadmap highlights the need to move from isolated model development toward fully integrated, clinically validated systems. Progression in data standardization,explainability, and multi-task learning will be a pathway for the next generation of dental AI research.*

16. Conclusion

This survey reviewed 47 deep learning studies published between 2021 and 2026, addressing dental diagnostics using panoramic and bitewing radiographs, covering caries detection, periodontal bone loss assessment, and related tasks . Our findings indicate that deep learning has reached clinically competitive levels across these sub tasks: caries detection achieves 96% recall with Attention U-Net [16]; periodontal landmark detection reaches mAP50 = 0.995 with YOLOv8 [4]; dual-disease classification achieves AUC $>$ 98% on periapical X-rays with EfficientNet-B0 and Grad-CAM [24]; and AI-assisted caries detection proves economically favorable over unaided clinical assessment at ICER = -13.9 €/year [25].

A closer look at the corpus reveals five structural gap.Most models focus on single-disease detection, overlooking the reality that dental diseases frequently co-occur.Automated severity staging, the most treatment-relevant output a diagnostic system could provide – remains almost absent from current pipelines, despite its centrality to clinical decision making.Explainability, though widely discussed, appears in only about 6.4% of the reviewed studies.In addition to this, dataset is heavily fragmented, taken from private, single-center collections with limited generalizability and minimal clinical integration into real clinical workflows

To our knowledge, no existing system simultaneously addresses caries detection, severity staging, periodontal bone loss quantification, and explainability within a single unified pipeline on panoramic radiographs. This represents the definitive unmet need identified by the reviewed literature. The individual technical components are available – YOLO detection, U-Net segmentation, Grad-CAM explainability, EfficientNet classification, and transformer-based global attention – have all been demonstrated on dental radiographic data. The remaining challenge is integration, clinical validation, and standardization: consolidating them into systems that are not merely accurate, but clinically trustworthy, interpretable, and actionable. This is essential for the junior clinicians and dental students who stand to benefit most from AI-assisted diagnostic support.

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